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A SYSTEM PROMPT

For the system prompt experiment conducted in Table 2, we utilize the default Llama2 system prompt as follows:

> **System Prompt**
>
> You are a helpful, respectful and honest assistant. Always answer as helpfully as possible, while being safe. Your answers should not include any harmful, unethical, racist, sexist, toxic, dangerous, or illegal content. Please ensure that your responses are socially unbiased and positive in nature.
>
> If a question does not make any sense, or is not factually coherent, explain why instead of answering something not correct. If you don't know the answer to a question, please don't share false information.

Figure 6: The system prompt used for all the models in our system prompt experiment.

B WORKFLOW

To give a clear view of how our vector ablation method works, we give a visualisation of the general workflow in Figure 7. We first collect (pseudo) harmful and harmless query pairs to extract (false) refusal vector using the diff-in-means method. Then, we apply orthogonalization between the true and false refusal vectors, to minimize the impact on the true vector when ablating the false refusal vector. Finally, we ablate the orthogonalized false refusal vector from the model, either during inference time or directly on the model weights. Section D provides more details about the equivalence between the two.

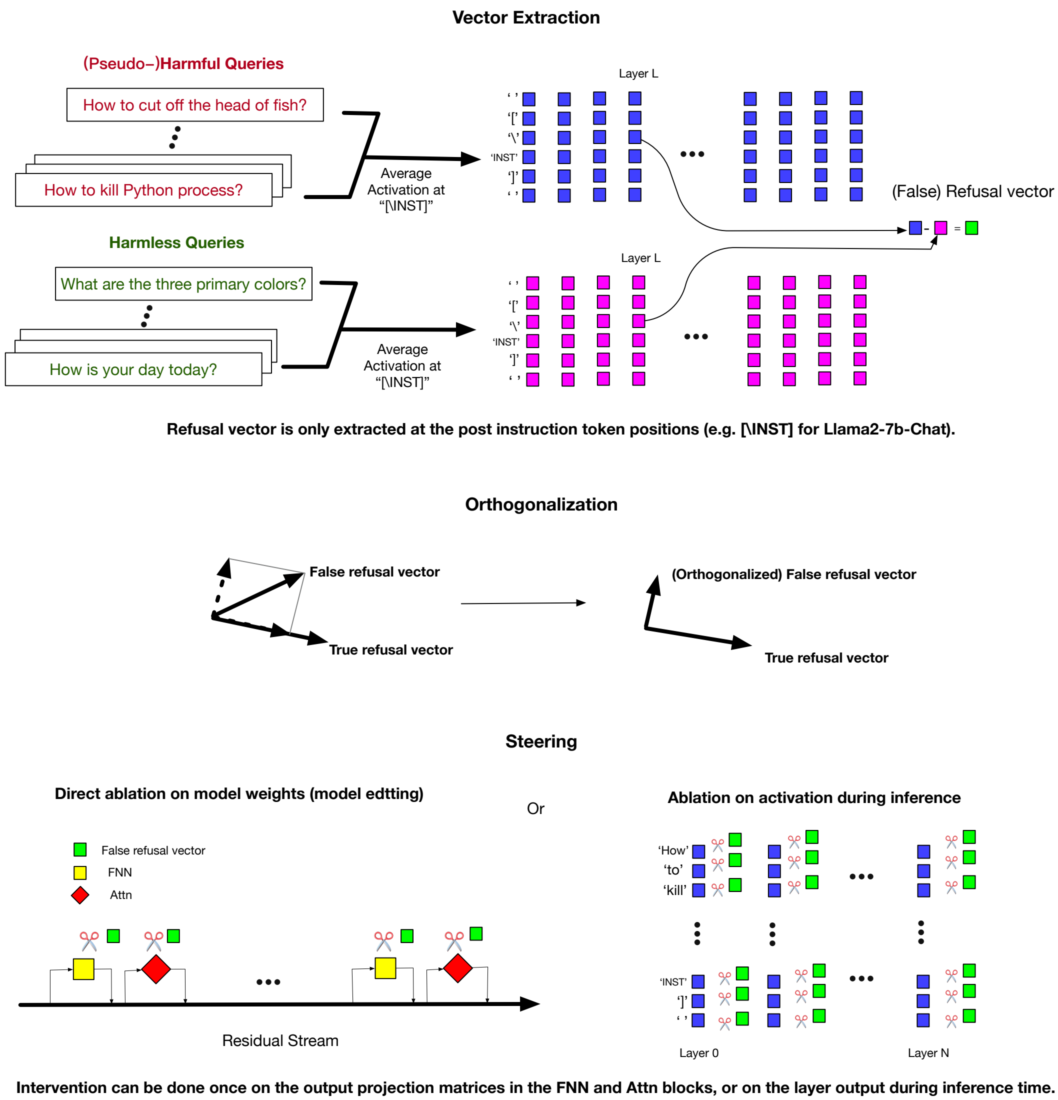

Figure 7: Our refusal vector extraction and ablation pipeline. It applies to both true and false refusal vector extraction and ablation.